Department of Computer Science and Engineering

Date: 12/01/2026	Test - 3	Max. Marks:60
Semester: 5	UG	Duration: 2 Hrs
Course Title: DATABASE MANAGEMENT SYSTEMS		Course Code: CD2521A

S. No.	Part A (Questions)	M	BT	CO
1	List and define the ACID properties of database transactions.	2	1	CO1
2	List the three conditions to identify that the two Transactions are conflicting to each other.	2	1	CO1
3	Mention the importance of Commit and Rollback.	2	2	CO2
4	Distinguish between Master-Slave replication and Peer-to-Peer replication.	2	1	CO1
5	Differentiate between SQL and NoSQL database.	2	2	CO3

S. No.	Part B (Questions)	M	BT	CO
1	Consider the relation $R = (A_1, A_2, A_3, A_4, A_5)$ and the set of functional dependencies $FD = \{A_1 \rightarrow A_3A_5,\ A_5 \rightarrow A_1A_4,\ A_3A_4 \rightarrow A_2\}$. Determine whether the decomposition $D = \{R_1, R_2, R_3\}$, where $R_1 = \{A_1, A_2, A_3, A_5\}$, $R_2 = \{A_1, A_3, A_4\}$, and $R_3 = \{A_4, A_5\}$, is a lossless join decomposition. Show all the necessary steps as per the algorithm.	10	4	CO3
2	Define concurrency. Explain the need for concurrent execution. List and describe the various problems encountered due to concurrent execution with relevant examples.	10	2	CO2

3	a) What is transaction? Explain the states of transaction execution with a neat diagram.	5	2	CO1
	b) Consider the following schedule of transactions.	5	2	CO3

T1	T2	T3
		Read(A)
	Read(A)	
		Write(A)
Read(A)		
Write(A)		

i. Construct the precedence graph for the given schedule.
ii. Check whether the above schedule is conflict-serializable or not. If it is conflict-serializable, write the equivalent serial schedule.

4	a) Explain the Column-Family Stores aggregate data model with a neat diagram.	5	3	CO3
	b) Differentiate between sharding and replication. Explain how sharding and replication can be combined, with a neat diagram.	5	2	CO1
5	a) Classify digital data with a neat diagram. Differentiate between structured, semi-structured, and unstructured data.	5	2	CO1
	b) Define Hadoop. Explain the key considerations of Hadoop with a neat diagram.	5	2	CO1